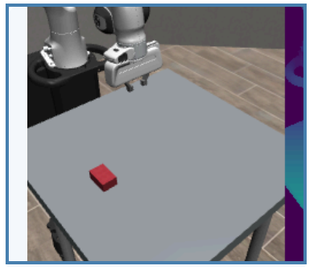
FORECAST

DB0 CORE: CLOSED-LOOP DYNAMIC MANIPULATION

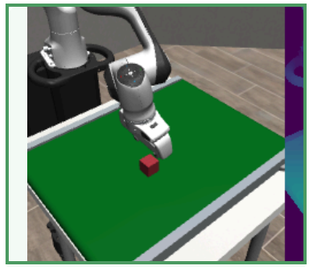
PHYSICS + EVENT STRESS



P0

Future Pose

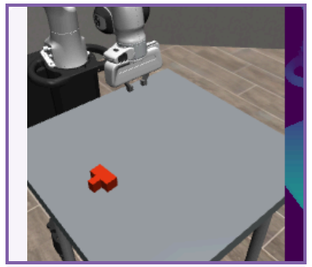
8 poses, 50-400 ms



T1-BELT

Intercept

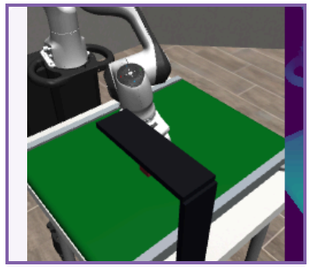
grasp + lift + hold



T1-XYZ / SO3

Orientation

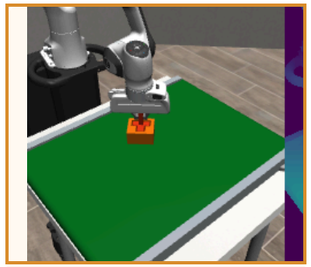
matched centroid motion



T1-OCC

Occlusion

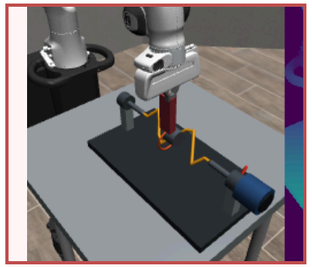
physical hidden interval



T2-TRANS / SE3

Moving Goal

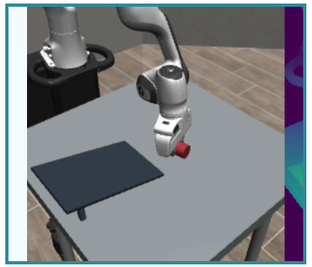
release + settle + dwell



T3-PHASE / FULL

Buzz Wire

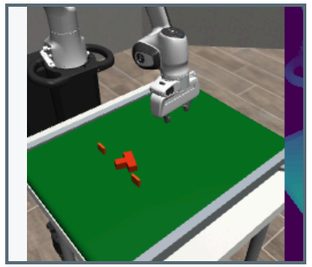
phase align, no touch



T4-SPHERE / SLIDER / CAN

Free Physics

roll / slide / slip



T5-REPLAN

Impact Replan

observe + correct + grasp

Isolated factors

- observability — current vs. valid history
- orientation — same XYZ, independent yaw
- visibility — physical occlusion interval
- time-to-contact — speed x delay x TTC
- free dynamics — gravity, friction, contact
- unannounced event — impact then re-observe

ONE PHYSICAL CLOCK

20 Hz RGB-D and control | 500 Hz physics | sensing, inference, actuation, contact, and dwell all consume time

PAIRED REPORTS

forecast | SR | TC | CT | safety | latency